GENERAL - LEGEND AND SYMBOLS

EQUIPMENT AND VALVE SYMBOLS


GATE VALVE


GLOBE VALVE


BALL VALVE


CHECK VALVE


CONTROL VALVE, DIAPHRAGM



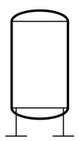
CONTROL VALVE, MOTOR

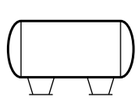

PRESSURE RELIEF VALVE


STRAINER


PIPE REDUCER


CENTRIFUGAL PUMP

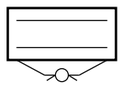

VERTICAL VESSEL / TANK

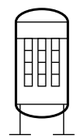

HORIZONTAL VESSEL / DRUM

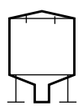


AGITATED JACKETED REACTOR

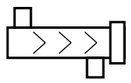

SHELL AND TUBE EXCHANGER

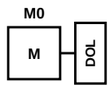

AIR COOLED EXCHANGER


CARTRIDGE FILTER


SILO / BULK STORAGE


FEED HOPPER


SCREW CONVEYOR



MOTOR WITH DOL STARTER

LINE SYMBOLS

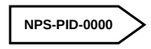
MAJOR PROCESS LINE

SECONDARY / UTILITY LINE

ELECTRIC SIGNAL

PNEUMATIC SIGNAL

CAPILLARY / FILLED SYSTEM



OFF PAGE CONNECTOR

INSTRUMENT SYMBOLS



FIELD MOUNTED INSTRUMENT



LOCAL PANEL MOUNTED



SHARED DISPLAY / CONTROL



INTERLOCK OR TRIP, LOGIC SOLVER

INSTRUMENT TAG CONVENTION

FIRST LETTER MEASURED VARIABLE
F FLOW L LEVEL P PRESSURE T TEMPERATURE A ANALYSIS
SUCCEEDING LETTERS READOUT OR FUNCTION
I INDICATE R RECORD C CONTROL T TRANSMIT V VALVE
Y COMPUTE A ALARM S SWITCH E ELEMENT Q TOTALISE
MODIFIERS H HIGH L LOW HH HIGH HIGH LL LOW LOW
LOOP NUMBER AREA PREFIX PLUS SEQUENCE
E.G. FIC-3001 IS AREA 300, LOOP 01, FLOW INDICATING CONTROLLER
CONTROL VALVES FV TAG ON A LEADER, SIGNAL TO THE ACTUATOR, FAIL ACTION BELOW
MOTOR DRIVES UX INTERLOCK BALLOON, DOL STARTER, MOTOR TAGGED Mn

NOTES

1. THIS DRAWING SET IS SYNTHETIC. IT DESCRIBES NO REAL FACILITY.
2. SYMBOLS FOLLOW THE CONVENTIONS OF ISA-5.1 UNLESS NOTED OTHERWISE.
3. ALL LINE SIZES ARE NOMINAL INCHES. ALL PIPING IS CARBON STEEL (CS)
4. UNLESS THE LINE NUMBER SUFFIX STATES OTHERWISE.
5. INSTRUMENT LOOP NUMBERS CARRY THE AREA NUMBER AS THEIR PREFIX.

DRAWING INDEX

NPS-PID-0001	LEGEND, SYMBOLS AND DRAWING INDEX
NPS-PID-0002	AREA 200 - RAW MATERIAL RECEIVING AND STORAGE
NPS-PID-0003	AREA 300 - FEED TRANSFER AND METERING
NPS-PID-0004	AREA 400 - REACTOR FEED PREHEAT
NPS-PID-0005	AREA 500 - PRIMARY REACTOR
NPS-PID-0006	AREA 600 - SEPARATION AND VAPOUR RECOVERY
NPS-PID-0007	AREA 700 - PRODUCT COOLING AND STORAGE
NPS-PID-0008	AREA 800 - UTILITIES, COOLING WATER AND STEAM

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

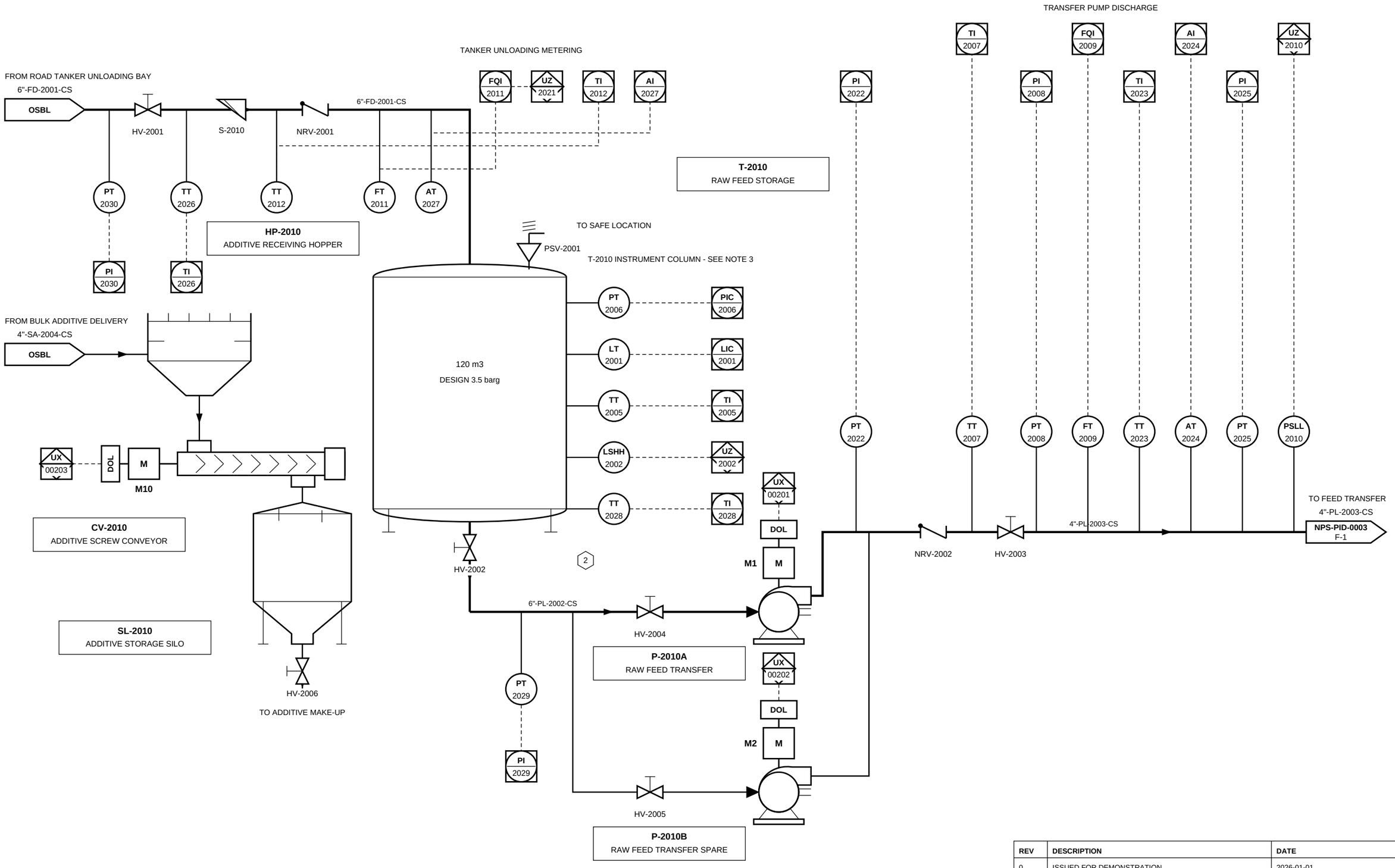
LEGEND, SYMBOLS AND DRAWING INDEX
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0001	**0**

SHEET	SCALE	DATE
01 OF 08	**NONE**	**2026-01-01**

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

AREA 200 - RAW MATERIAL RECEIVING AND STORAGE



NOTES

1. TANK T-2010 IS BLANKETED WITH NITROGEN AT 20 mbarg.
2. SUCTION LINE TO BE LAID WITH A CONTINUOUS FALL TOWARDS THE PUMPS.
3. LSHH-2002 TRIPS THE UNLOADING PUMP AND CLOSES HV-2001.
4. P-2010A AND P-2010B ARE IDENTICAL. B IS AN INSTALLED SPARE ON AUTO START.
5. CV-2010 IS INTERLOCKED TO STOP ON HIGH LEVEL IN SL-2010.

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

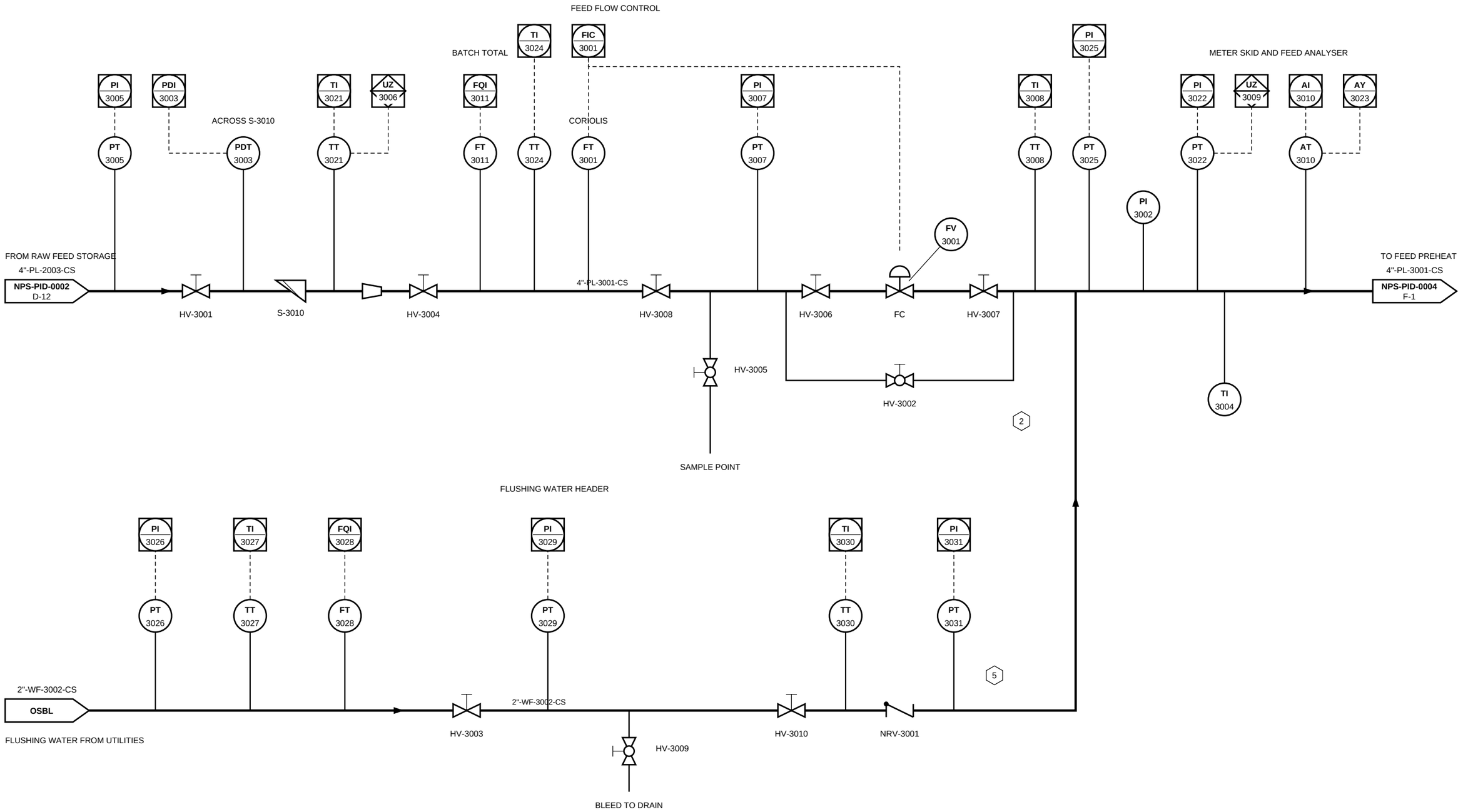
AREA 200
RAW MATERIAL RECEIVING AND STORAGE
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0002	**0**

SHEET	SCALE	DATE
02 OF 08	**NONE**	**2026-01-01**

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

AREA 300 - FEED TRANSFER AND METERING



NOTES

1. FT-3001 IS A CORIOLIS MASS METER. INSTALL WITH 10D STRAIGHT RUN UPSTREAM.
2. MANUAL BYPASS HV-3002 IS CAR SEALED CLOSED.
3. FV-3001 FAILS CLOSED ON LOSS OF INSTRUMENT AIR OR SIGNAL.
4. S-3010 IS FITTED WITH A DIFFERENTIAL PRESSURE GAUGE.
5. FLUSHING WATER IS ISOLATED AND BLINDED DURING NORMAL OPERATION.
6. AT-3010 SAMPLES THE METERED FEED. A HIGH RESULT STOPS THE TRANSFER.

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

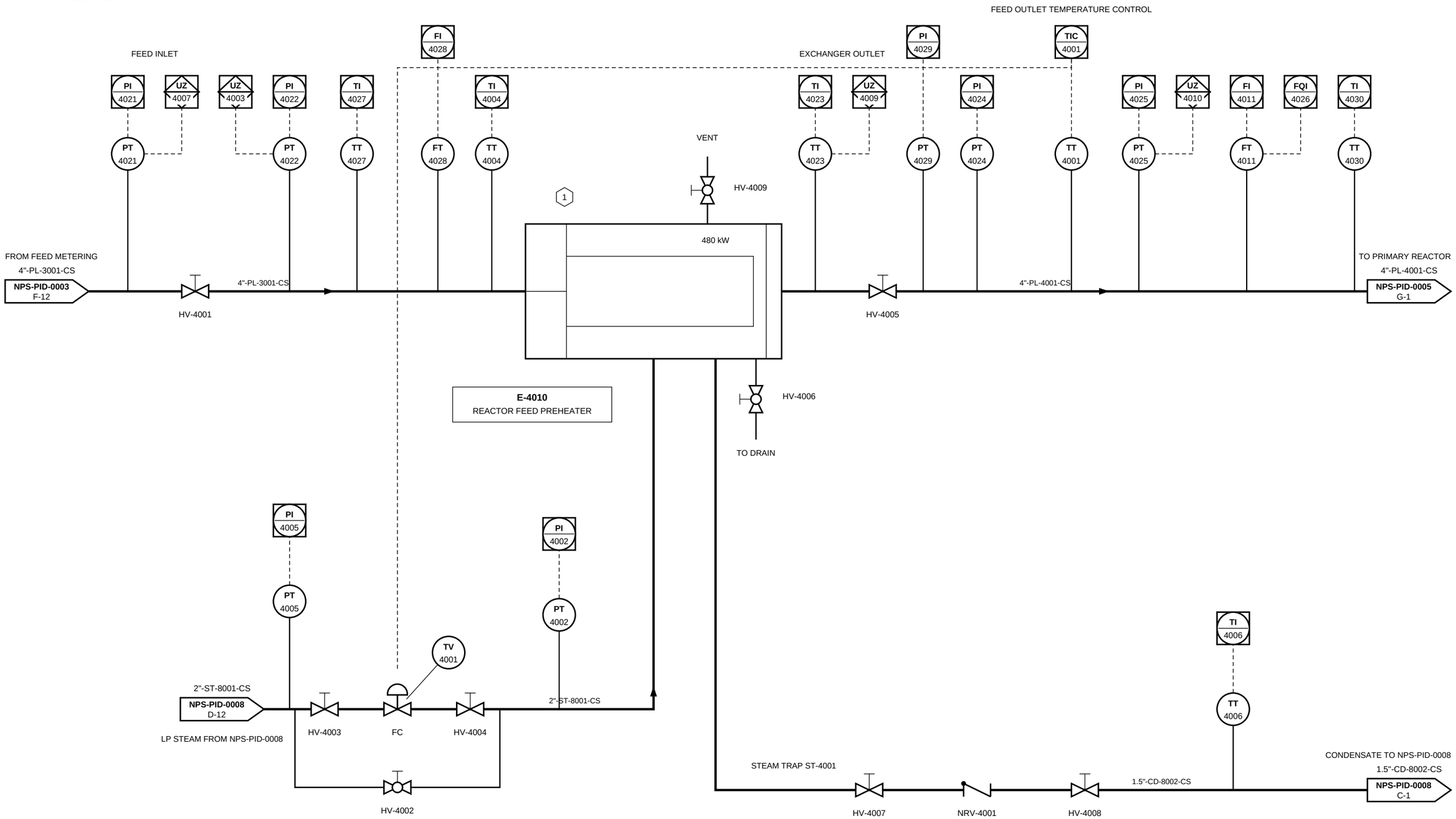
NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

AREA 300
FEED TRANSFER AND METERING
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0003	**0**

SHEET	SCALE	DATE
03 OF 08	**NONE**	**2026-01-01**

AREA 400 - REACTOR FEED PREHEAT



NOTES

1. E-4010 IS A TWO PASS SHELL AND TUBE UNIT, FEED ON THE TUBE SIDE.
2. TIC-4001 THROTTLES LP STEAM TO HOLD 85 degC AT THE FEED OUTLET.
3. CONDENSATE RETURNS TO THE FLASH RECEIVER ON SHEET 08.
4. TV-4001 FAILS CLOSED.

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

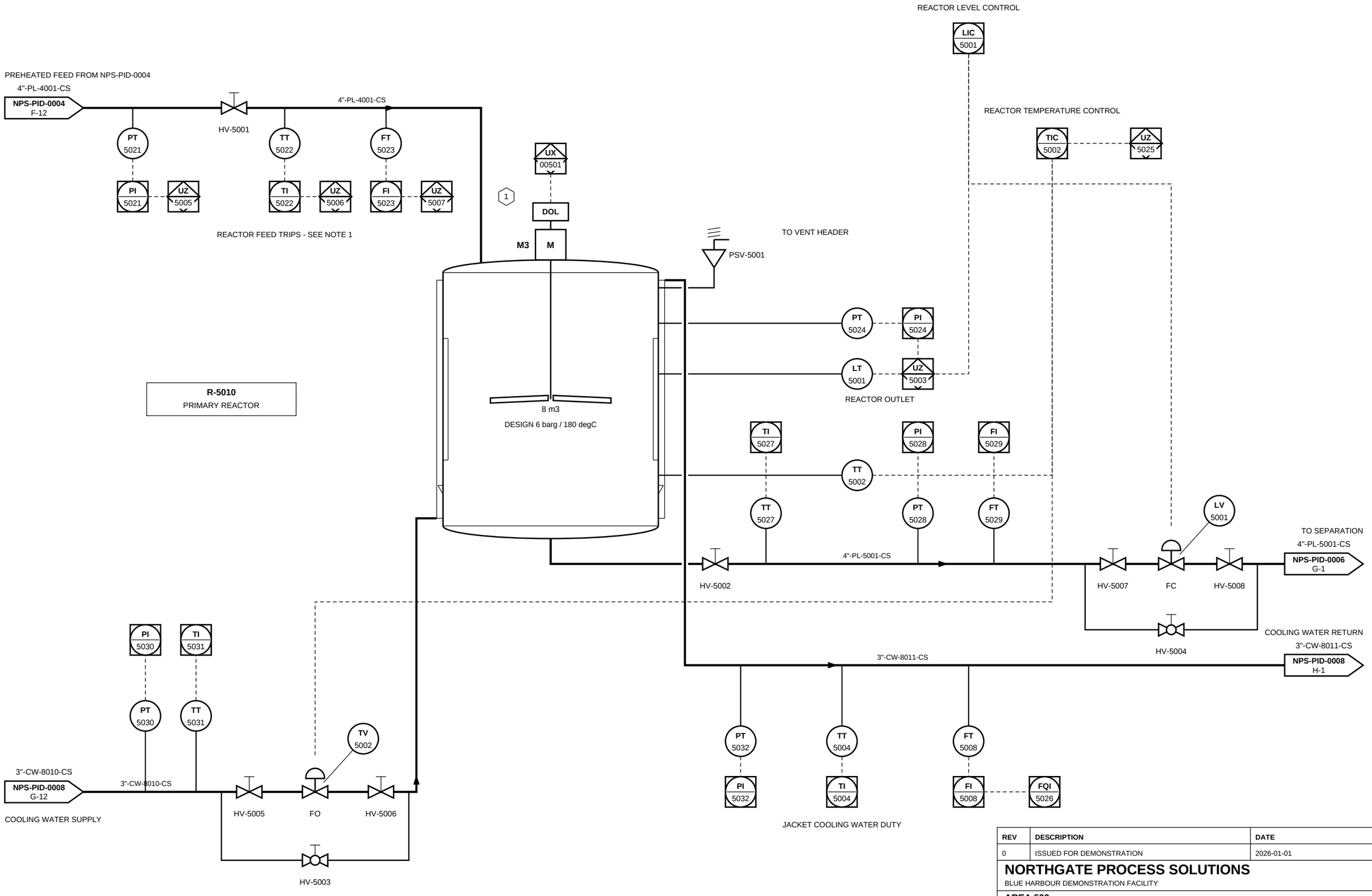
AREA 400
REACTOR FEED PREHEAT
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0004	**0**

SHEET	SCALE	DATE
04 OF 08	**NONE**	**2026-01-01**

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

AREA 500 - PRIMARY REACTOR



NOTES

1. R-5010 IS AGITATED CONTINUOUSLY. AGITATOR FAILURE TRIPS THE FEED VALVE.
2. TV-5002 FAILS OPEN TO MAINTAIN COOLING ON LOSS OF SIGNAL.
3. LV-5001 HOLDS REACTOR LEVEL AT 60 PERCENT.
4. PSV-5001 IS SET AT 6 barg AND DISCHARGES TO THE VENT HEADER.

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

AREA 500
PRIMARY REACTOR
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0005	**0**

SHEET	SCALE	DATE
05 OF 08	**NONE**	**2026-01-01**

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

AREA 600 - SEPARATION AND VAPOUR RECOVERY

DRUM PRESSURE CONTROL

OVERHEAD VAPOUR MONITORING

PI 6026 | PIC 6001 | PI 6028 | PI 6021 | UZ 6005 | FI 6006 | FQI 6022 | TI 6023 | UZ 6007 | AI 6008 | UZ 6024 | TI 6027

PT 6021 | FT 6006 | TT 6023 | PT 6026 | PT 6001 | AT 6008 | TT 6027 | PT 6028

FROM PRIMARY REACTOR

4"-PL-5001-CS

NPS-PID-0005
D-12

HV-6001

2

4"-PL-5001-CS

PV 6001

VAPOUR TO RECOVERY UNIT

6"-VP-6001-CS

OSBL

HV-6010

6"-VP-6001-CS

HV-6004 | FO | HV-6005

PI 6003 | PT 6003

PSV-6001

HV-6002

DRUM LEVEL CONTROL

LIC 6002

4.5 m3

DESIGN 4 barg

LT 6002

HV-6009

V-6010
PRODUCT FLASH DRUM

TI 6029 | PI 6009 | TI 6025 | FQI 6030 | PI 6031

TT 6029 | PT 6009 | TT 6025 | FT 6030 | PT 6031

HV-6008

LV 6002

TO PRODUCT COOLING

3"-PL-6002-CS

3"-PL-6002-CS

NPS-PID-0007
F-1

HV-6006 | FC | HV-6007

TI 6004 | TT 6004

HV-6003

NOTES

1. V-6010 OPERATES AT 1.2 barg AND 95 degC.
2. PIC-6001 SPLIT RANGE IS NOT USED. PV-6001 FAILS OPEN.
3. LV-6002 IS SIZED FOR 120 PERCENT OF NORMAL BOTTOMS FLOW.
4. PSV-6001 DISCHARGES TO THE VENT HEADER ON SHEET 08.

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

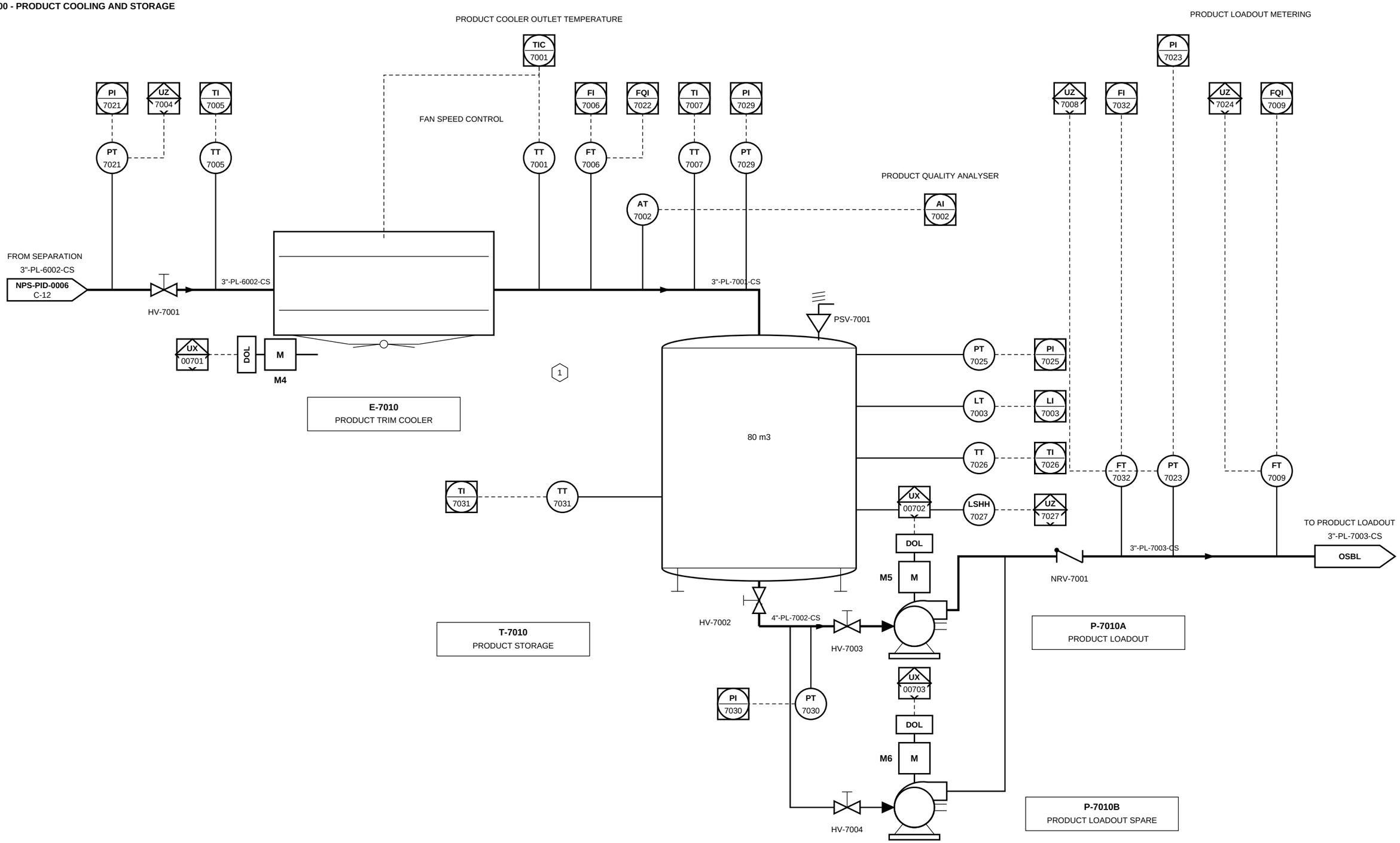
NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

AREA 600
SEPARATION AND VAPOUR RECOVERY
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0006	0

SHEET	SCALE	DATE
06 OF 08	NONE	2026-01-01

AREA 700 - PRODUCT COOLING AND STORAGE



NOTES

1. E-7010 FAN SPEED IS MODULATED BY TIC-7001 TO HOLD 40 degC.
2. AT-7002 SAMPLES CONTINUOUSLY. HIGH RESULT DIVERTS TO OFF SPEC STORAGE.
3. T-7010 IS NITROGEN BLANKETED.
4. P-7010A AND P-7010B ARE IDENTICAL. B IS AN INSTALLED SPARE.

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

AREA 700
PRODUCT COOLING AND STORAGE
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0007	**0**

SHEET	SCALE	DATE
07 OF 08	**NONE**	**2026-01-01**

AREA 800 - UTILITIES, COOLING WATER AND STEAM

COOLING WATER CIRCUIT

CW RETURN FROM AREA 500

3"-CW-8011-CS

NPS-PID-0005 C-12

3"-CW-8011-CS

TT 8004 | PT 8005 | FSLL 8006

TI 8004 | PI 8005 | UZ 8006

CW RETURN HEADER

UX 00801 | DOL | M

M7

TI 8026

TT 8026

E-8010
COOLING WATER COOLER

6"-CW-8012-CS

HV-8006

P-8010A
COOLING WATER CIRCULATION

UX 00802 | DOL | M

M8

CW HEADER PRESSURE

PIC 8001

PT 8001

6"-CW-8010-CS

NRV-8001

HV-8001

CW SUPPLY TO AREA 500

3"-CW-8010-CS

NPS-PID-0005 B-1

PSLL 8011 | FT 8012 | TT 8021

UZ 8011 | FQI 8012 | TI 8021

UX 00803 | DOL | M

M9

CW SUPPLY HEADER

FQI 8025

FT 8025

HV-8007

P-8010B
COOLING WATER CIRCULATION SPARE

LP STEAM AND CONDENSATE

PI 8023 | TI 8007 | FQI 8008 | UZ 8009 | TI 8010 | PI 8024

PT 8023 | TT 8007 | FT 8008 | PSLL 8009 | TT 8010 | PT 8024

MP STEAM FROM SITE HEADER

3"-ST-8000-CS

OSBL

HV-8002

3"-ST-8000-CS

FLASH VAPOUR TO LP HEADER

PV 8002

HV-8004 | FC | HV-8005

HV-8003

LP STEAM TO NPS-PID-0004

2"-ST-8001-CS

NPS-PID-0004 C-3

PI 8027 | TI 8028

PT 8027 | TT 8028

1.5"-CD-8002-CS

NPS-PID-0004 B-12

1.5"-CD-8002-CS

CONDENSATE FROM NPS-PID-0004

LT 8003 | LI 8003

TI 8029

TT 8029

2

PT 8002

PIC 8002

LP STEAM HEADER - LETDOWN TO 3 barg

2"-CD-8003-CS

OSBL

CONDENSATE TO SITE RETURN

V-8010
CONDENSATE FLASH RECEIVER

NOTES

1. COOLING WATER IS TREATED CLOSED CIRCUIT. MAKE UP IS MANUAL.
2. PV-8002 LETS MP STEAM DOWN FROM 10 barg TO 3 barg.
3. V-8010 FLASH VAPOUR IS RECOVERED INTO THE LP STEAM HEADER.
4. P-8010A AND P-8010B ARE IDENTICAL. B IS AN INSTALLED SPARE.

REV	DESCRIPTION	DATE
0	ISSUED FOR DEMONSTRATION	2026-01-01

NORTHGATE PROCESS SOLUTIONS
BLUE HARBOUR DEMONSTRATION FACILITY

AREA 800
UTILITIES - COOLING WATER AND STEAM
PROJECT NPS-0000

DRAWING NUMBER	REV
NPS-PID-0008	**0**

SHEET	SCALE	DATE
08 OF 08	**NONE**	**2026-01-01**

SYNTHETIC DRAWING - GENERATED FOR SOFTWARE DEMONSTRATION - NOT FOR CONSTRUCTION